

First-Rib Stress Fracture in Overhead Throwing Athletes

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Background: First-rib stress fracture is considered a potential cause of nonspecific atraumatic chronic shoulder pain in adolescent athletes. However, the etiology in throwing athletes with first-rib fracture is still unknown. The purpose of this retrospective study was to investigate the characteristic clinical features and radiographic findings of overhead throwing athletes with first-rib fracture.

Methods: Twenty-four first-rib stress fractures in 23 players were studied retrospectively. Clinical features, including age, initial symptom, sports, pain-related activity, diagnostic method, treatment method, and final follow-up, were reviewed.

Results: The following characteristic clinical features were identified: mean age of 16.8 years (range, 13 to 25 years), 19 dominant arm injuries and 5 non-dominant arm injuries, and an acute increase in pain while swinging the bat or pitching the ball. Sixteen fractures presented with posterior shoulder or upper thoracic back pain. At a mean time of 7.5 months after the initiation of conservative treatment, 17 healing fractures (71%) and 7 nonunion fractures (29%) among throwing athletes with first-rib stress fracture were identified. On image analysis, first-rib stress fractures were classified into 3 types depending on the direction and location of fracture lines: groove, intrascapular, and posterior types. Three symptomatic patients underwent first-rib resection due to thoracic outlet syndrome. On average, 46% of the first rib was visible on the shoulder radiographs and 97% was visible on the cervical spine radiographs. The Cohen kappa coefficient for the above percentages was 0.87 and the percent agreement was 89.4% for the shoulder, and the Cohen kappa coefficient was 0.80 and the percent agreement was 99.0% for the cervical spine.

Conclusions: First-rib stress fracture should be considered when adolescent overhead throwing athletes have acute-onset posterior shoulder pain while swinging the bat or pitching the ball. Anteroposterior radiography of the cervical spine is available for initial diagnosis. Although 71% of the patients healed at a mean follow-up of 7.5 months with conservative treatment, some patients may have symptoms consistent with thoracic outlet syndrome.

Level of Evidence: Therapeutic Level IV. See Instructions for Authors for a complete description of levels of evidence.

First-rib fracture is very rare¹. A previous study showed 73 first-rib fractures in 77,607 consecutive radiographs of servicemen¹. Several case reports have confirmed that first-rib fracture may be associated with high-energy direct impact, indirect trauma, and sports activities²⁻⁴. Fractures of the first rib are uncommon in athletes and present a different clinical entity from traumatic first-rib fracture associated with high-energy thoracic trauma⁵. First-rib stress fracture is a potential cause of nonspecific atraumatic chronic shoulder pain in adolescent overhead throwing athletes^{4,6-14}. These studies showed involvement of the dominant and nondominant sides in baseball players. Curran and Kelly reported a first-

rib stress fracture in the nondominant arm of a professional baseball player⁶.

Symptoms are usually nonspecific and include diffuse pain in the region of the shoulder girdle⁵. Although shoulder pain is one of the most common symptoms in baseball players, diagnosis of first-rib stress fracture is difficult. Therefore, the number of patients with first-rib stress fracture may be larger than expected. Barrett et al. pointed out that this injury may be overlooked as a cause of shoulder pain in athletes¹⁵. Thus, the etiology of first-rib stress fracture in throwing athletes is still unknown. The purpose of this retrospective study was to investigate characteristic

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clinical features and available radiographs of overhead throwing athletes with first-rib fracture.

Materials and Methods

Patients

Data from consecutive patients who were treated in our hospitals were retrospectively reviewed. This study was approved by the relevant institutional review board. Between August 2002 and June 2017, 38 patients were diagnosed with first-rib stress fracture. The diagnosis of first-rib stress fracture was based on radiography or computed tomography (CT). The inclusion criteria were symptomatic first-rib stress fracture and no obvious traumatic episode in throwing athletes. The exclusion criteria were follow-up of <1 year, participation in other overhead throwing sports, and being a non-athlete. Eight patients with follow-up of <1 year and 7 nonthrowing athletes were excluded from this study. Twenty-four first-rib stress fractures in 23 players were studied retrospectively (Table I). Clinical features, including age, initial symptom, sports, pain-related activity, diagnostic method, treatment method, and final follow-up, were reviewed. Two orthopaedic surgeons investigated fracture patterns and union rate on radiographic evaluation.

Availability in Diagnostic Radiograph

The percentage of the first rib visible on radiographs was assessed in age-matched controls for whom either an anteroposterior shoulder radiograph or a cervical spine radiograph was available. Two orthopaedic surgeons assessed 2 different radiographs for availability as a diagnostic tool: 67 anteroposterior radiographs of the shoulder in patients with pain related to throwing motion and 102 anteroposterior radiographs of the cervical spine in patients presenting with neck pain.

Statistical Analyses

The interobserver reliability of the above measurement was assessed on the shoulder radiographs and on the cervical spine radiographs with use of the Cohen kappa statistic and the rate of agreement between the judgments of the 2 observers. All statistical data were analyzed using R version 2.8.1 (The R Project for Statistical Computing [CRAN freeware]).

Results

The following characteristic clinical features were identified: mean age of 16.8 years (range, 13 to 25 years), 19 dominant arm injuries and 5 non-dominant arm injuries, and an acute increase in pain while swinging the bat ($n = 9$) or pitching the ball ($n = 6$). Seventeen cases presented with pain at the posterior aspect of the shoulder, inferior or medial border of the scapula, and upper thoracic region of the back. Other symptoms were painful breathing in 5 patients. At a mean time of 7.5 months after the initial treatment, 17 healing cases (71%) and 7 nonunion cases (29%) of throwing athletes with first-rib stress fracture were identified. Of the 7 nonunion cases, 3 cases were symptomatic, and the patients underwent first-rib resection

because of thoracic outlet syndrome. During the follow-up period, 1 patient (case 18) had a recurrence of first-rib stress fracture and 1 patient (case 12) had a symptomatic thoracic outlet syndrome after healing of the first rib.

Radiographic and CT analysis revealed 3 classifications of first-rib stress fractures based on the direction and portion of fracture lines: groove type (3 cases), in which the fracture line was along the subclavian artery groove; intrascapular type (18 cases), in which the fracture line was inside the middle scalene muscle insertion; and posterior type (3 cases), in which the fracture was next to the costovertebral articulation (Fig. 1).

Representative Cases

Case 11

A 20-year-old, left-hand-dominant, male professional baseball pitcher presented with periscapular pain on the right side (nondominant side) for several weeks. Although the back pain decreased after 2 months, he reported numbness of the right upper extremity when he pitched. He presented to our hospital after conservative treatment for several months. Radiography and CT demonstrated a groove-type first-rib stress fracture. The patient's clinical presentation, positive provocative tests, and narrowing of the subclavian artery on 3-dimensional angiography led to the diagnosis of thoracic outlet syndrome with nonunion of a first-rib fracture. The fracture line was along the subclavian artery groove (Figs. 2-A, 2-B, and 2-C). The patient underwent a transaxial partial resection of the first rib. After 2 months postoperatively, he was able to pitch, and after 6 months postoperatively, he returned to his previous level of play.

Case 13

A 14-year-old, right-hand-dominant, male baseball player who hits right-handed presented with right elbow pain on the dominant side. The pain had been improving over the last 3 months despite no clear diagnosis. The pain subsequently worsened, and, for 1 year, he had dull pain and numbness in the right elbow and upper extremity. Radiography and CT revealed an intrascapular type of a nonunion first-rib fracture. The fracture line was remote from the position and direction of the subclavian artery (Figs. 2-D, 2-E, and 2-F). The patient underwent a transaxial partial resection of the first rib. The patient had excellent outcomes, and no obvious complications, including neurovascular disturbance, were observed.

Availability in Diagnostic Radiograph

The presence of a first rib was 46% in shoulder radiographs and 97% in cervical spine radiographs (Fig. 3). For the shoulder, the Cohen kappa coefficient was 0.87 and the rate of agreement was 89.4%. For the cervical spine, the Cohen kappa was 0.80 and the rate of agreement was 99.0%.

Discussion

Shoulder pain is one of the most common symptoms in baseball players¹⁶⁻¹⁸. The pathology of the shoulder pain often involves the glenohumeral joint, such as a labral tear,

TABLE I Demographic Data*

Case	Age† (yr)	Sex	Radiographs at First Examination	Visibility of First Rib on Shoulder Anteroposterior View	Injured Side	Throwing	Batting	Sports	Position
1	14	M	Chest	Possible	R	R	NA	Tennis	NA
2	14	M	Cervical spine	Possible	L	R	Bilateral	Baseball	Pitcher
3	16	M	Chest	Possible	R	R	R	Baseball	Fielder
4	17	M	Cervical spine and shoulder	Cervical spine possible; shoulder impossible	R	R	R	Baseball	Pitcher
5	21	M	Shoulder	Impossible	R	R	R	Baseball	Fielder
6	15	M	Chest and cervical spine	Possible	R	R	R	Baseball	Fielder
7	17	M	Rib	Unclear	R	R	R	Baseball	Fielder
8	16	M	Rib	Unclear	L	R	R	Baseball	Fielder
9	15	M	Shoulder	Impossible	L	L	L	Baseball	Pitcher
10	15	M	Shoulder	Possible	R	R	R	Baseball	Fielder
11	20	M	Chest	Possible	R	L	L	Baseball	Pitcher
12	16	M	Rib	Possible	R	R	R	Baseball	Fielder
13	14	M	Rib	Possible	R	R	R	Baseball	Fielder
14	14	M	Rib	Unclear	R	R	R	Baseball	Fielder
15	13	M	Cervical spine	Possible	R	R	R	Baseball	Pitcher
16	16	M	Shoulder	Impossible	R	R	R	Baseball	Fielder
17	18	M	Rib	Possible	R	R	L	Baseball	Fielder
18	19	M	Rib	Possible	L	R	L	Baseball	Fielder
19	17	M	Chest	Impossible	R	R	R	Baseball	Pitcher
20	25	F	Shoulder	Impossible	R	R	R	Baseball	Pitcher
21	23	M	Shoulder	Possible	L	R	R	Baseball	Fielder
22	18	M	Acromioclavicular joint	Impossible	L	R	R	Javelin throw	NA
23	16	M	Rib	Possible	R	R	NA	Baseball	Pitcher
24	15	M	Rib	Impossible	R	R	R	Baseball	Fielder

*NA = not applicable. †The mean patient age was 16.83 years. ‡The mean time to healing was 7.5 months. §The mean time to return to sport was 3.4 months. #The mean follow-up was 37.0 months.

rotator cuff tear, or capsular ligament problem. Therefore, radiographs of the shoulder joint are usually the focus of diagnosis. However, other pathologies of the shoulder joint should be considered for appropriate diagnosis of cases outside of the glenohumeral joint, such as impingement syndrome, muscle fatigue, laxity and resultant shoulder instability, thoracic outlet syndrome, and first-rib stress fracture. When no clear etiology of the shoulder pain and back pain among baseball players can be identified, first-rib stress fracture should be considered. The current study demonstrated the characteristic clinical features and radiographic findings of overhead throwing athletes with a first-rib fracture.

This study showed that characteristic clinical features of the first-rib stress fracture were adolescent age, back or posterior shoulder pain, and acute onset of pain when swinging the bat or throwing. Previous studies have demonstrated that most patients with first-rib stress fracture sustain shoulder pain located under the scapula and posterior to the clavi-

cle^{9,19,20}. Therefore, these symptoms may mistakenly lead to concerns regarding the presence of shoulder pathology. Because symptoms mimic those of ipsilateral shoulder injury, some patients with first-rib stress fracture have been misdiagnosed, which can lead to pseudarthrosis.

Nonspecific findings secondary to acute-onset posterior shoulder pain are frequently the starting points for the workup and evaluation for the first-rib stress fracture in overhead throwing athletes (Fig. 4). In addition, characteristic physical examination findings could not be identified in this study. We consider that nonspecific findings are characteristics of a first-rib stress fracture. Therefore, this absence of characteristic physical examination finding may lead to a difficulty in diagnosis. In a retrospective review of patients with blunt chest trauma, Luceri et al. identified 56 patients with first-rib fractures, only 20% of which were seen on a chest radiograph²¹. In the current series, an anteroposterior radiograph of the shoulder provided an accurate diagnosis for only 1 case (1 of 6) at the initial examination. An

TABLE I (continued)

Primary Symptom	Onset	Treatment	Radiographic Type	Healing	Time to Healing† (mo)	Return to Sport§ (mo)	Follow-up# (mo)	Outcomes After Return to Sport
Painful breathing	Unclear	Conservative	Posterior	Union	2.8	2.0	142.2	Satisfactory
Posterior shoulder pain	Swinging bat	Conservative	Intrascapular	Union	1.4	1.4	24.0	Elbow pain
Painful breathing	Unclear	Conservative	Intrascapular	Union	0.9	0.9	76.6	Satisfactory
Posterior shoulder pain	Swinging bat	Conservative	Intrascapular	Union	2.1	2.1	14.9	Elbow pain
Posterior shoulder pain	Pitching	Conservative	Intrascapular	Union	4.0	3.2	12.1	Retired
Posterior shoulder pain	Unclear	Conservative	Intrascapular	Union	3.2	3.2	123.8	Satisfactory
Breathing	Unclear	Conservative	Groove	Union	8.5	3.5	65.6	Shoulder pain
Painful breathing	Swinging bat	Conservative	Groove	Nonunion	NA	2.4	113.3	Satisfactory
Posterior shoulder pain	Swinging bat	Conservative	Intrascapular	Union	5.6	2.1	12.0	Satisfactory
Posterior shoulder pain	Swinging bat	First-rib resection	Intrascapular	Resection	NA	2.0	24.0	Satisfactory
Painful breathing	Pitching	First-rib resection	Groove	Resection	NA	2.0	16.4	Satisfactory
Posterior shoulder pain	Swinging bat	Conservative	Intrascapular	Union	3.2	3.2	39.0	Thoracic outlet syndrome treated conservatively
Elbow pain	Unclear	First-rib resection	Intrascapular	Resection	NA	2.0	24.5	Satisfactory
Posterior shoulder pain	Pitching	Conservative	Intrascapular	Union	3.9	2.3	12.0	Satisfactory
Posterior shoulder pain	Swinging bat	Conservative	Intrascapular	Union	7.5	5.0	22.8	Satisfactory
Posterior shoulder pain	Unclear	Conservative	Posterior	Nonunion	NA	1.2	12.2	Satisfactory
Posterior shoulder pain	Pitching	Conservative	Intrascapular	Nonunion	NA	6.4	18.0	Retired
Posterior shoulder pain	Pitching	Conservative	Intrascapular	Union	3.4	3.4	38.2	Recurrence
Posterior shoulder pain	Swinging bat	Conservative	Intrascapular	Union	4.3	4.3	13.3	Good
Posterior shoulder pain	Pitching	Conservative	Intrascapular	Nonunion	NA	4.2	24.0	Shoulder pain
Posterior shoulder pain	Swinging bat	Conservative	Posterior	Union	16.8	14.0	14.0	Satisfactory
Posterior shoulder pain	Unclear	Conservative	Intrascapular	Union	3.8	3.8	13.9	Satisfactory
Shoulder pain	Unclear	Conservative	Intrascapular	Union	7.8	5.7	19.0	Satisfactory
Posterior shoulder pain	Unclear	Conservative	Intrascapular	Union	1.7	1.6	12.6	Satisfactory

anteroposterior radiograph of the shoulder was not valuable for the diagnosis of the first-rib stress fracture, in contrast to an anteroposterior radiograph of the cervical spine.

We compared age-matched shoulder and cervical spine radiographs for diagnosis of first-rib stress fracture. The percent availability was 46% for shoulder radiographs and 97% for cervical spine radiographs. According to the angle of the radiographs, the preferred visible angle of the first rib would be similar to Luschka (uncovertebral) joints at the central region as one of the indicators of a standard cervical spine radiograph²². The angulation of the costovertebral joints has been shown to increase from C5 to C7²³. Therefore, the visibility of these joints in the central region of the cervical spine was an indicator that the first rib would also be visible on that radiograph. We considered that cervical spine radiography can be a diagnostic tool of first-rib stress fracture. Adolescent overhead throwing sports players with acute-onset back or posterior shoulder pain, in particular, should be evaluated for first-rib stress fracture via cervical radiography.

Stress fractures result from excessive, repetitive, submaximal loads on bones that cause an imbalance between bone resorption and formation²⁴. Excessive use or a rapid increase in training have been suggested to contribute to stress fracture of the first rib in athletes⁴. Several pathologies have been suggested to explain the etiology of first-rib stress fractures, including weakness of the first rib, strength imbalance, and muscle fatigue. Stress fractures of the first rib commonly occur at the groove for the subclavian artery, just posterior to the scalene tubercle^{6,19,25}. The anatomy of the first rib makes it particularly susceptible to stress fracture. The deeper subclavian artery groove, immediately posterior to the scalene tubercle and between the attachments of the scalenus anterior and medius muscles, appears to be the weakest point in the rib because most stress fractures occur at this location^{6,26}. The anterior scalene muscle pulls the lateral aspect of the first rib posterolaterally, and the middle scalene muscle pulls the medial aspect of the rib in a medial direction. The upward pull of both of the scalene muscles opposes the antagonistic downward pull of the serratus anterior and intercostal muscles and traction of

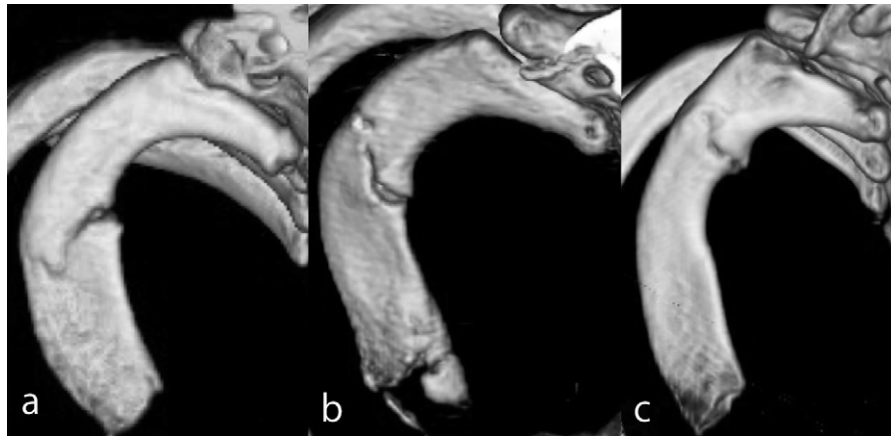


Fig. 1

Analysis of radiographs and CT images revealed that first-rib stress fractures were classified into 3 types depending on the direction and portion of the fracture lines: groove type (**Fig. 1-A**), in which the fracture line is along the subclavian artery groove; interscalene type (**Fig. 1-B**), in which the fracture line is inside the middle scalene muscle insertion; and posterior type (**Fig. 1-C**), in which the fracture is beside the costovertebral articulation.

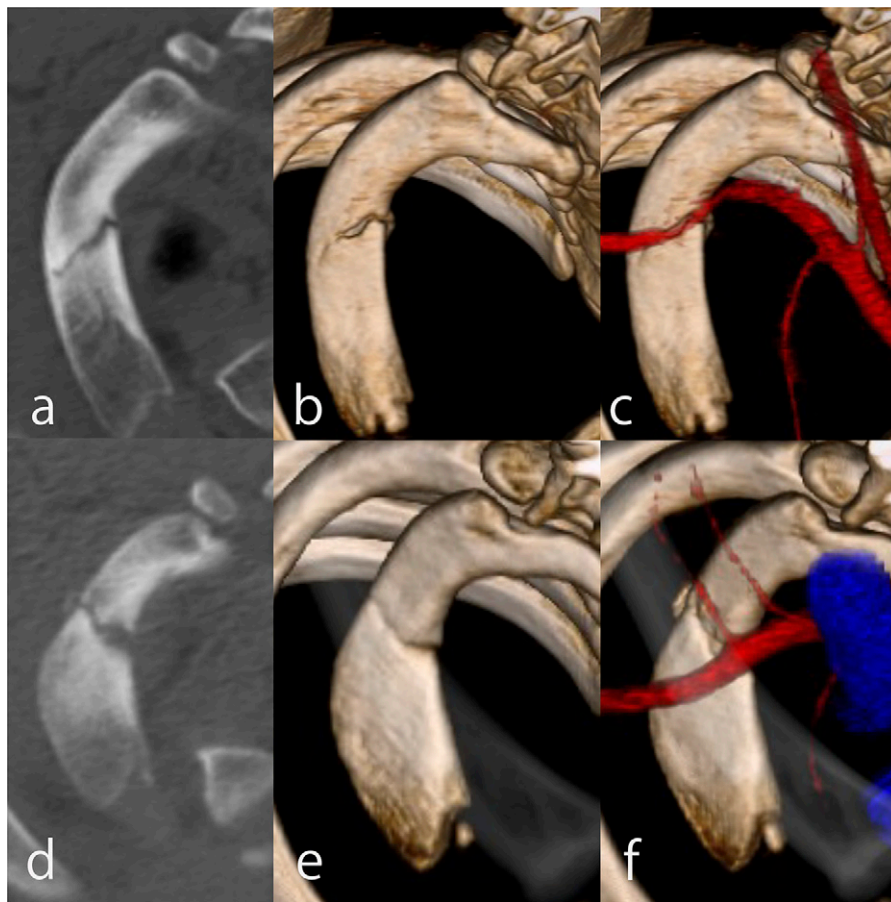


Fig. 2

Figs. 2-A through 2-F Representative cases. **Figs. 2-A, 2-B, and 2-C** Case 11. A 20-year-old, left-hand-dominant, male professional baseball pitcher presented with periscapular pain on the right (nondominant) side. **Fig. 2-A** CT image showing a nonhealing fracture line. **Fig. 2-B** Three-dimensional CT image showing a slightly displaced medial border of the first rib without bridging callus. **Fig. 2-C** Three-dimensional angiography image of the fracture line along the subclavian artery. **Figs. 2-D, 2-E, and 2-F** Case 13. A 14-year-old, right-hand-dominant, male baseball player who hits right-handed presented with elbow pain on the right (dominant) side. **Fig. 2-D** CT image showing a nonhealing fracture line. **Fig. 2-E** Three-dimensional CT image showing slight separation of the fracture line. **Fig. 2-F** Three-dimensional angiography image showing that the fracture line has a different portion and direction than the subclavian artery.

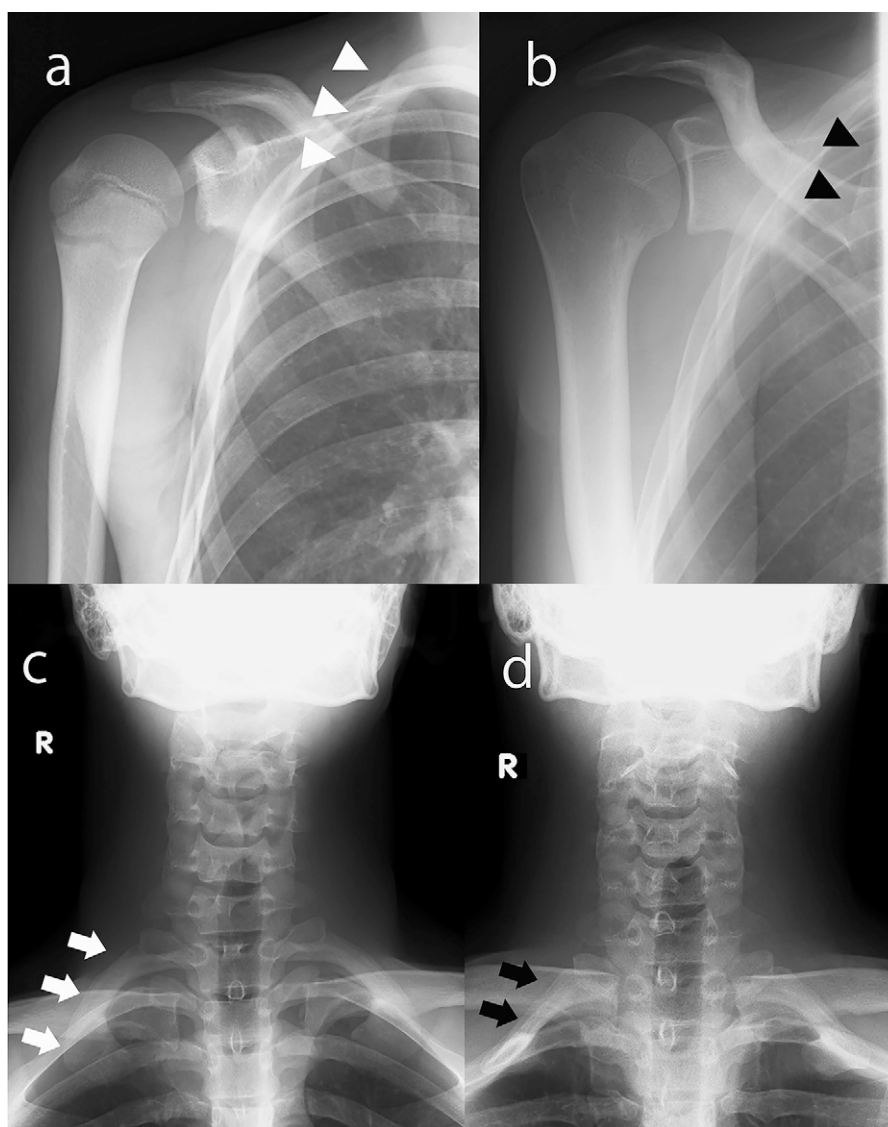


Fig. 3

Figs. 3-A through 3-D Radiographs of the shoulder and cervical spine of age-matched controls. The percent availability for the first rib was 46% on the shoulder radiograph and 97% on the cervical spine radiograph. **Fig. 3-A** Anteroposterior view of the shoulder. White arrowheads show the visible first rib. **Fig. 3-B** Anteroposterior view of the shoulder. Black arrowheads show the non-visible portion of the first rib. **Fig. 3-C** Anteroposterior view of the cervical spine. White arrows show the visible first rib. **Fig. 3-D** Anteroposterior view of the cervical spine. Black arrows show the non-visible portion of the first rib.

the arm⁹. Radiographic and CT analysis revealed 3 classifications of first-rib stress fractures: groove type (12.5%), intrascapular type (75%), and posterior type (12.5%). These types may indicate various pathologies of first-rib stress fracture. In case 11, the fracture line was consistent with the location of the subclavian artery as shown by angiography. These findings with regard to the 3 fractures as classified by groove type suggest that the subclavian artery groove is the weakest mechanical point of the first rib. In contrast, the position of the fracture line in case 13 differed from that of the subclavian artery seen on 3-dimensional angiography. When resection of the first rib was performed for symptomatic thoracic outlet syndrome, the

fracture line was found to be located in the middle of the scalene muscle (i.e., an intrascapular-type fracture). Therefore, different pathologies may cause different types of first-rib stress fracture. As described, the etiology of stress fractures is multifactorial, including a direct blow, muscle contractions, and a repetitive bending force to the lateral segment of the rib^{9,25,27}. Biomechanical testing may be able to confirm the existence of several mechanisms for first-rib stress fractures.

Treatment of a rib stress fracture is typically conservative. Almost all of these injuries heal spontaneously without any intervention after 4 to 6 weeks of rest²⁰. However, previous articles and the present results showed an unexpectedly high

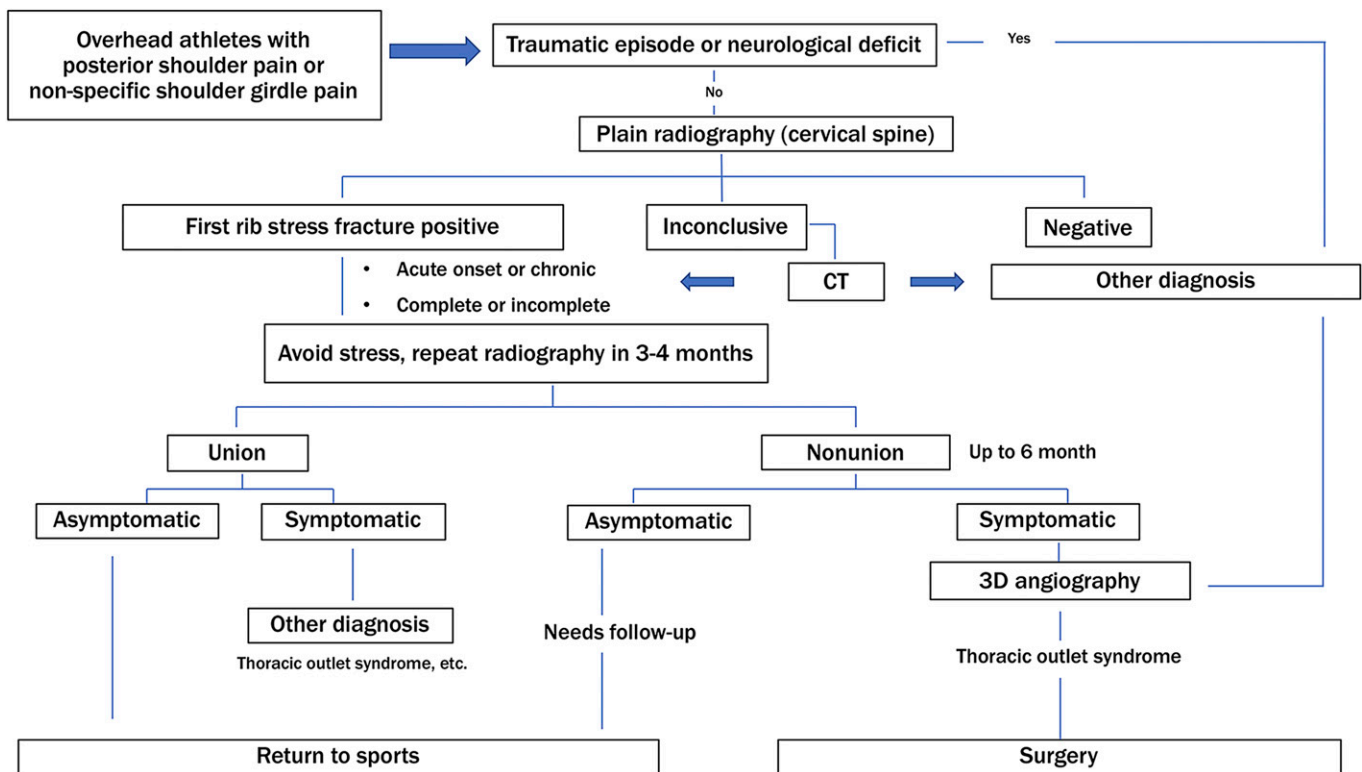


Fig. 4

An algorithm for the investigation and treatment of a first-rib stress fracture. 3D = 3-dimensional.

rate of pseudarthrosis of first-rib stress fractures^{5,8}. Nonunion of complete fractures has been documented, but it may not result in obvious symptoms or affect return to play¹⁰. The present study demonstrated that although the union rate was 71% and patients can return to sports at 3 months, symptomatic thoracic outlet syndrome or recurrence of the stress fracture should be considered. Most patients with pseudarthrosis were able to play without any symptoms around the shoulder and back. On the contrary, the current series indicated that we should be mindful of the possibility of thoracic outlet syndrome in patients with pseudarthrosis of a first-rib stress fracture. Previous case reports showed that athletes may have thoracic outlet syndrome associated with nonunion of a first-rib fracture²⁸⁻³⁰. In our series, some patients with or without healing of the first-rib stress fracture had the symptoms of thoracic outlet syndrome. As previously noted, the pathology of a first-rib stress fracture is still unknown. However, because the region of the thoracic outlet is defined as being between the anterior and middle scalene muscles and between the clavicle and the first rib, thoracic outlet syndrome could reasonably be considered a risk if a stress fracture is located either between the anterior and middle scalene muscles or inside the scalene muscles. Therefore, the first-rib fracture would be considered to be associated with thoracic outlet syndrome. When neurovascular symptoms with or without healing of a first-rib stress fracture were found and were not improved after rehabilitation of scapulothoracic dysfunction, we recommended CT angiography to detect thoracic outlet syndrome.

Our study had limitations. The study group was small, and the follow-up was short. The symptom recurrence rate and union rate should be assessed using a large sample with longer follow-up. Given the small sample size, we could not compare union and complication rates according to the different fracture types. Moreover, the times required for healing and return to sports were inherently inaccurate because patients were not followed radiographically and clinically on a daily or weekly basis. Although we investigated the percentage availability of the visible first rib of the shoulder and cervical spine radiographs, the sensitivity and specificity of these radiographs in making the diagnosis could not be assessed because of the lack of consistent imaging. Lastly, sports that did not involve overhead throwing were not investigated, and the characteristic mechanism of first-rib stress fracture in overhead throwing athletes could not be determined in the current study. Further biomechanical study could clarify the mechanism of first-rib stress fractures and long-term prognosis, including union rate or rate of thoracic outlet syndrome, based on the different fracture types.

In conclusion, specific clinical features of the first-rib stress fracture were adolescent overhead throwing athletes, posterior shoulder pain, and acute onset of pain while swinging the bat or pitching the ball. An anteroposterior radiograph of the cervical spine was valuable for the initial diagnosis. Image analysis revealed 3 types of first-rib stress fractures, namely, groove, intrascapular, and posterior types. Although 71% of the patients in our study healed within a mean follow-up of 7.5

months with conservative treatment, some patients may have symptomatic thoracic outlet syndrome. The orthopaedic surgeon should be aware of the possibility of a first-rib stress fracture in an adolescent who plays an overhead throwing sport and has back or posterior shoulder pain after an acute event, to avoid misdiagnosis and the possible occurrence of a symptomatic nonunion. ■

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